Journal of Applied Physics

RESEARCH ARTICLE | AUGUST 04 2010

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J. Appl. Phys. 108, 033103 (2010)
https://doi.org/10.1063/1.3466996

AIP Publishing

Laser-induced damage threshold of silicon in millisecond, nanosecond, and picosecond regimes

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(Received 31 March 2010; accepted 24 June 2010; published online 4 August 2010)

Millisecond, nanosecond, and picosecond laser pulse induced damage thresholds on single-crystal are investigated in this study. The thresholds of laser-induced damage on silicon are calculated theoretically for three pulse widths based on the thermal damage model. An axisymmetric mathematical model is established for the transient temperature field of the silicon. Experiments are performed to test the damage thresholds of silicon at various pulse widths. The results indicate that the damage thresholds obviously increase with the increasing of laser pulse width. Additionally, the experimental results agree well with theoretical calculations and numerical simulation results. © *2010 American Institute of Physics*. [doi:10.1063/1.3466996]

I. INTRODUCTION

The study of laser–semiconductor interaction mechanism is a very active field over the years and attracts more interest recently. On the one hand, semiconductor is the basic material of photodetector which is apt to be destroyed in optics-electronic countermeasure. The study of damage mechanism of photodetector induced by laser is based on the study of laser–semiconductor interaction mechanism. On the other hand, laser can carry out the microprocessing of semiconductor which is widely used in the microelectronic field recently. Furthermore, with the development of photovoltaic solar industry, laser is widely used in cutting single-crystal silicon, polycrystalline silicon, and amorphous-silicon solar cell. And it is feasible to cut and to reshape the silicon wafer using the laser thermal effect which can avoid crack initiation effectually.[1]

The application fields above mentioned are based on the laser thermal effect which is one of the most important damage mechanisms during the process of laser–semiconductor interaction. The studies of the theory, mechanisms, measurement, and amelioration of laser-induced damage threshold (LIDT) have been investigated a lot in the past few years. Nevertheless, most of these studies have devoted to ultrashort laser pulse (femtosecond, picosecond pulse width), short laser pulse (nanosecond pulse width) and continuum laser. Up to now, very few studies relate to the damage of semiconductor induced by millisecond laser pulse.

LIDT of semiconductor is influenced by the material surface morphology, the laser wavelength, pulse width, the number of shots, etc.[2] Bonse *et al.*[3] studied the temporal dynamics of melting, ablation and detected the surface morphology, and the reflectivity using femtosecond-time-resolved microscopy, real-time streak camera and photodiode measurements upon femtosecond laser pulse irradiation of single-crystalline germanium. They indicated that the ablation threshold of the germanium was 0.83 $\mathrm{J/cm^2}$. Singh *et al.*[4] took into account the thermal and mechanical damage of GaAs in picosecond regime and carried out the analytical and experimental investigations. They found that the damage threshold of GaAs was 0.7 $\mathrm{J/cm^2}$. Iwata and Asakawa[5] examined accumulative damage of GaAs and Inp surfaces induced by 20 ns multiple-laser-pulse irradiation. They concluded that the damage threshold decreased when the sample was irradiated with a large number of pulses. Garg[6] pointed out that the above effects is observed only when the repetition rate was higher than 1 Hz and the damage threshold of GaAs induced by single shot laser was 0.9 $\mathrm{J/cm^2}$. Qi *et al.*[7] investigated the damage of a single-crystal GaAs wafer experimentally induced by a 1064 nm continuous laser by checking the intensity change in probe beam. A complete self-consistent model for transport dynamics in semiconductor caused by ultrashort pulse laser heating was presented by Chen *et al.*[8] They obtained the temporal and spatial evolution of the carrier density and temperature as well as the lattice temperature with a finite difference method and the damage threshold of silicon and germanium for pulse width 0.01–10 ps. Shen *et al.*[9] introduced a double-temperature model to calculate the heating process in silicon irradiated by picosecond and nanosecond lasers, respectively, with the same energy density on the basis of the absorption mechanism of the semiconductor. Thermal damage model in terms of the coupled diffusion equations was introduced by Meyer *et al.*[10,11] The physical processes involved in the conversion of high intensity optical energy to lattice heat have been considered for optical heating in semiconductors. They showed the damage thresholds of semiconductor for a broad range of laser wavelengths, among which the damage threshold of silicon for nanosecond laser was 2 $\mathrm{J/cm^2}$. Arora *et al.*[12] examined the damage of single-crystal silicon and silicon-based photodetectors irradiated by nanosecond laser based on Meyer's idea. They figured out that the damage threshold of silicon was 3.6 $\mathrm{J/cm^2}$ for a single shot laser, and obtained the value of damage threshold of silicon for multiple pulses varying laser repetition frequency.

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The studies mentioned above are not related to the millisecond laser. There are some distinctive features for millisecond laser, such as huger energy and longer irradiation time than nanosecond and femtosecond laser during the interaction process. Moreover, air breakdown will hardly occur and no plasma is generated during the long-pulse laser irradiation. It is well known that plasma will absorb the laser energy and become opaque to the laser beam. Thus, the energy waste is much smaller for long-pulse laser than short-pulse laser during the long-distance transmission process. It is convenient to destroy the detector at a far distance by long-pulse laser. Hence, in present study we perform a detailed investigation of LIDT of silicon for millisecond, nanosecond, and picosecond laser. Theoretical calculations in terms of material heating depth are performed. Furthermore, an axisymmetric mathematical model is established for the laser–silicon interaction. The numerical solution of temperature field is obtained by using the finite element method in the above regimes. Experiments are conducted to verify the theoretical and numerical results. It is shown that the theoretical calculations and the numerical simulations are consistent with the experimental results. Besides, comparing the damage threshold for three pulse widths, we find that the damage threshold will increase with increasing of laser pulse width.

II. THEORETICAL CALCULATION

A. Absorption mechanism

There are four main absorption mechanisms during the process of laser–semiconductor interaction depending on laser parameters and material characters. The first is intrinsic absorption whose absorption coefficient is of the order of $10^5-10^6\ \text{cm}^{-1}$. Electron–hole pairs are generated when electrons are thermally excited from valence band into conduction band. The second is exciton absorption. It is negligible in the room temperature because that the exciton can only be detected in very low temperature due to low ionization energy. The third is free-carrier absorption in which the carriers absorb the energy and the energy is transferred to lattice via the interaction. The last is impurity absorption which is the dominant absorption mechanism only when the material is adulterated heavily. Consequently, intrinsic absorption and free-carrier absorption are the two crucial factors in the laser–silicon interaction.

As mentioned previously, the laser energy transition process in silicon is listed as follows: the generation of carriers, carriers absorb energy, and the energy is transferred from carrier to lattice. The incident laser wavelength is 1064 nm and the corresponding photon energy $h\nu = h(c/\lambda) = 1.17$ eV. The band gap E_g of silicon is 1.12 eV at room temperature 300 K. Thus, the following relationship is established: $h\nu > E_g$. As an indirect band semiconductor, silicon absorbs the laser energy and takes interband transition. Meanwhile the nonequilibrium carriers generated by impact ionization and the equilibrium carriers absorb the laser energy and exchange energy via relaxation collision. The energy is transferred to lattice by coupling with lattice. The temperature rises and reaches the fusion and boiling points due to the energy deposition.

B. Damage threshold

The incident laser energy density E is required to rise the temperature of the material surface ΔT, which can be theoretically estimated by the relation:[11]

$$E = \frac{\rho(T)c(T)L_H\Delta T}{1-R(T)}, \tag{1}$$

where $\rho(T)$, $c(T)$, and $R(T)$ are the density of the material, specific heat, and reflectivity. The thermal properties ρ, c, and R are presently assumed to be independent of the temperature to simplify the calculation. L_H is the material heating depth which is given by:[10]

$$L_H = \frac{1-R_0}{c_0\Delta T}\int_{T_0}^{T_f}\frac{c}{\alpha(1-R)} \times \frac{1}{\dfrac{\chi_T}{L_T\alpha+1}+\dfrac{\chi_B^{NR}}{L_T\alpha+L_D\alpha+1}+\dfrac{\chi_S^{NR}}{L_T\alpha}}dT. \tag{2}$$

Rearrangement of Eq. (2) gives:

$$L_H = \frac{1}{\Delta T}\int_{T_0}^{T_f}\frac{1}{\alpha}\times\frac{1}{\dfrac{\chi_T}{L_T\alpha+1}+\dfrac{\chi_B^{NR}}{L_T\alpha+L_D\alpha+1}}dT, \tag{3}$$

where α is the total optical absorption coefficient, L_T is the thermal diffusion depth, and L_D is the free-carrier diffusion depth. The parameters χ_T, χ_B^{NR}, χ_S^{NR} are the fractions of the laser energy entering the sample which go, respectively, into the thermalization of excess carriers immediately after excitation, into nonradiative bulk recombination, and into nonradiative surface recombination.

Assuming that the initial sample temperature is the room temperature T_0, the melting point is T_f, and damage is due to surface melting. Therefore, when the center of beam reaches the melting point of the material, the damage threshold E_{th} can be defined:[11]

$$E_{th} = \frac{\rho(T)c(T)L_H(T_f-T_0)}{1-R(T)}. \tag{4}$$

For 1 ms laser, the pulse width is much longer than the carrier diffusion time. There is enough time for electrons to transfer excess energy to lattice and the thermal equilibrium state will be established instantaneously. The laser energy is converted to heat which is transferred to lattice via the coupling between carrier and lattice. Once the absorbed energy is converted to lattice heat, it is free to diffuse to greater depths via thermal conduction. Therefore, the thermal diffusion length L_T plays an important role in determining the heating depth. $\chi_T \approx 0.5$, $\chi_B^{NR} \approx 0.5$,[11] Eq. (3) can be written as:

$$L_H = \frac{1}{\alpha} + L_D + L_T, \tag{5}$$

where $L_T = \sqrt{(k\tau/\rho c)}$, in which τ is pulse width. $L_T >> \alpha^{-1} + L_D$.Thereby, $\alpha^{-1} + L_D$ can be negligible when calculation approximately. The following function is obtainable from Eqs. (4) and (5) accordingly:

$$E_{th} = \frac{\rho c(T_f - T_0)}{1 - R}\sqrt{\frac{k\tau}{\rho c}}. \tag{6}$$

It can be concluded from Eq. (6) that the damage threshold of silicon is proportional to $\sqrt{\tau}$ for long-pulse laser. The thermal properties of silicon are taken as follows:[13] mass density $\rho = 2.33$ g/cm^3, specific heat $c = 0.721$ J/(g K), thermal conductivity $k = 1.05$ W/(cm K), and reflectivity $R = 0.33$.The pulse width $\tau = 0.001$ s. We obtained $L_T = 250$ μm. Herein, we obtain the damage threshold of silicon is 124 J/cm^2 for 1 ms pulse width laser evaluated from Eq. (4).

For 10 ns pulse laser, the peak value of carrier density is 1.3×10^{21} cm^{-3}, when it exceeds 10^{20} cm^{-3} carrier effects are taken into consideration.[9] Auger electrons rapidly thermalize and transfer the energy to the lattice. Since the carrier collision time in the energy transfer process is of the order of 10^{-11} to 10^{-13} s, carriers reach a quasiequilibrium state in 10 ns. While the electron and hole recombine and the energy is transferred to new carrier. The temperature of the carrier and the lattice will rise simultaneously. Heat transfer it's too late happen for nanosecond pulses, therefore, the thermal diffusion depth L_T will be neglected. Thus, for ns pulse L_H is written as:

$$L_H = \frac{1}{\alpha} + L_D, \tag{7}$$

where $\alpha = \alpha_1 + \alpha_2 + \alpha_{FC}$, α_1 is the one-photon band-to-band absorption coefficient, α_2 is the two-photon band-to-band absorption coefficient, α_{FC} is the free-carrier absorption coefficient. $L_D = \sqrt{D\tau_B}$, D is the ambipolar diffusion coefficient, τ_B is the bulk free-carrier lifetime. When irradiated by 1064 nm laser, the band-to-band absorption coefficient of silicon is 11 cm^{-1}, and α_{FC} is as much as 10^3 cm^{-1} which is much greater than band-band absorption,[12] α_2 is much smaller than α_1 and α_{FC} which can be negligible. $L_H \approx 10$ μm.[14] We insert the value of L_H into Eq. (4), and the damage threshold for 10 ns pulse laser is 5 J/cm^2.

For 10 ps laser, the newly created electron and hole exchange energy through collisions with other carriers. Since thermalization typically occurs on a picosecond time scale, significant carrier diffusion has not yet occurred. Thus, carrier diffusion length can be neglected for picosecond laser. Most of energy can reach the lattice after electron–hole recombination, which induced that the temperature of lattice is lower a few times than the carriers. Only a little energy transfer to the lattice during the picosecond laser irradiation, most of the energy deposited in the carrier system which can transferred to lattice via relaxation process after laser irradiation. Hereby, the material heating depth for the picosecond laser can be depicted as:[10]

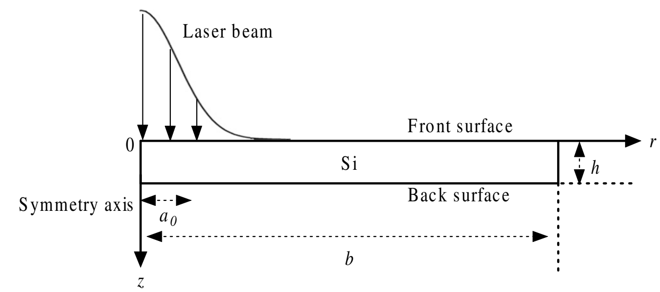

FIG. 1. Schematic diagram of the silicon irradiated by laser.

$$L_H = \frac{1}{\alpha}, \tag{8}$$

where $L_H \approx 2.47$ μm. Evaluated from Eq. (4), the damage threshold decreases significantly to 0.73 J/cm^2 for 10 ps pulse laser.

III. NUMBERICAL SIMULATION

A. Mathematical model

The incident laser beam is aligned perpendicularly to the substrate surface and assumed to be Gaussian distribution. A two-dimensional finite element model is established as shown in Fig. 1, and in which cylindrical coordinate system is adopted. Axis z is the symmetry axis, r shows the direction of radius and O is the central point on the front surface. The depth and radius of the sample are h and b, respectively. To make the problem manageable, some assumptions are made as follows: (1) the material is homogeneous and isotropic; (2) the model is large enough that heat energy will not transfer to the boundary during laser irradiation; (3) temperature of the back surface is supposed to be thermal insulation;(4) thermal losses by convection and radiation of the material surface are neglected; and (5) no taking account of the phase of silicon after vaporization.

It must be recognized that limit to damage by thermal mechanisms alone occurs at pulse lengths longer than 10^{-13} s, multiphoton avalanche ionization gradually become dominant at pulse lengths shorter than this.[15] Therefore, it is feasible to simulate the heating process of silicon based on thermal conduction theory for 10 ps–1 ms pulse width. The thermal conduction equation can be described as:[16]

$$\rho c\frac{\partial T(r,z,t)}{\partial t} = \frac{1}{r}\frac{\partial}{\partial r}\left(rk\frac{\partial T(r,z,t)}{\partial r}\right) + \frac{\partial}{\partial z}\left(k\frac{\partial T(r,z,t)}{\partial z}\right) + q, \tag{9}$$

where $T(r,z,t)$ represents the temperature distribution at time t, q is the heat source, which is described as:

$$q = I_0(1 - R)\alpha f(r)g(t)\exp(-\alpha z). \tag{10}$$

The boundary conditions is:

$$-k\left.\frac{\partial T(r,z,t)}{\partial z}\right|_{z=h} = -k\left.\frac{\partial T(r,z,t)}{\partial r}\right|_{r=b} = 0, \tag{11}$$

where I_0 is the incident laser peak power density at the spot center, R and α represent the reflectivity and absorption coefficient of the material surface, respectively. Besides, no

TABLE I. Thermophysical properties for silicon.

The thermal physical parameter	Solid state	Liquid state
ρ(g cm^{-3})	2.33	2.53
α(cm^{-1})	50	8.6×10^5
c[J/(g K)]	0.72	1.02
k[W/(cm K)]	$299/(T-99)$	0.23
R	0.33	0.72

heat flux transfers along the radial direction on the z-axis in the axisymmetric model.

$$-k\left.\frac{\partial T(r,z,t)}{\partial r}\right|_{r=0}=0, \tag{12}$$

the initial condition is:

$$T(r,z,t)|_{t=0}=T_0, \tag{13}$$

where T_0 is the ambient temperature and it is taken as 300 K.

The spatial model of the incident laser beam is assumed as Gaussian distribution, $f(r)$ and $g(t)$ is the spatial and temporal distribution of the laser pulse. These two functions for TEM_{00} mode laser pulse can be written as:

$$f(r)=\exp\left(-\frac{r^2}{a_0^2}\right),$$

$$g(t)=1,$$

$$I(r,t)=I_0\exp\left(-\frac{2r^2}{a_0^2}\right), \tag{14}$$

where a_0 is beam radius ($1/e^2$ of the intensity).

B. Material parameters

The initial silicon substrate temperature is 300 K, and the model is $1\times 10\ \text{mm}^2$. The thermal parameters including the density, absorption coefficients, the specific heat, the thermal conductivity are listed in Table I.[13,17] Most of the parameters change with the temperature. Plasma will be generated during nanosecond and picosecond pulse width laser irradiation which become opaque to the laser beam. Meanwhile, the shock wave induced by plasma expansion will damage the material surface. Therefore, plasma effect will be considered as absorption coefficient to modify the simulation of nanosecond and picosecond laser. The temperature keeps monotonically during the solid–liquid, liquid–vapor phase transition process, in which the absorbed or the emitted energy is called phase change latent heat. The phase change latent heat is considered as the enthalpy of material, the enthalpy is written as: $H=\int\rho C(T)dT$. The latent heats of silicon for melting and vaporization is 1800 J g^{-1} and 16 207 J g^{-1}, respectively.[13]

C. Finite element method

According to Eqs. (9)–(13), the FEM controlling equation for temperature can be written as:[18,19]

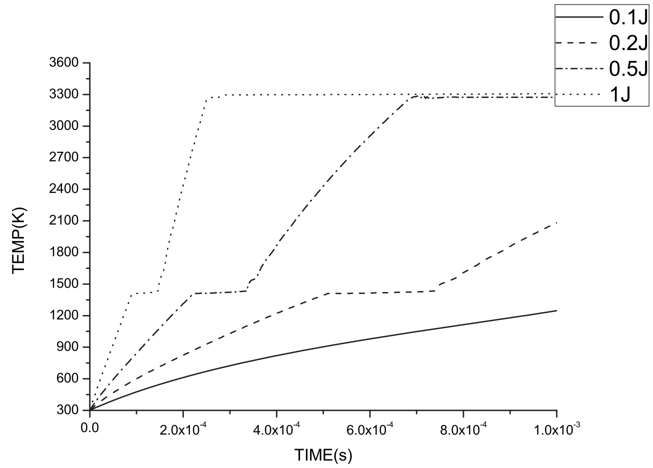

FIG. 2. Temperature evolution of the spot center for 1 ms laser.

$$[C]\{\dot{T}\}+[K]\{T\}=\{Q\} \tag{15}$$

where $[C]$, $\{\dot{T}\}$, $[K]$, $\{T\}$ are the heat capacity matrix, temperature rise rate vector, the heat conduction matrix, and temperature vector, respectively. The absorbed laser energy is expressed by the heat source vector $\{Q\}$, $\{\dot{T}\}=d\{T\}/dt$. Crank–Nicolson method is used to solve Eq. (15), and the temporal discrete form of Eq. (15) can be derived:

$$\left[\frac{1}{\Delta t}(C)+\theta(K)\right]\{T\}_t=\left[\frac{1}{\Delta t}(C)-(-\theta)(K)\right]\{T\}_{t+\Delta t}+\theta\{Q\}_t+(1-\theta)\{Q\}_{t+\Delta t}, \tag{16}$$

where Δt is the time interval, and $\theta=1/2$.

D. Numerical results

In present study, we calculate the temperature field of silicon for 1 ms, 10 ns, and 10 ps laser, respectively. The temperature evolutions at the center of laser irradiation under different laser energy are illustrated in Figs. 2–4.

It can be seen from Fig. 2 that, for 1 ms pulse width laser, the temperature rise of silicon is 1246 K when the laser energy is 0.1 J and the corresponding laser energy density is 79.6 J/cm^2. The peak temperature is up to about 2081 K

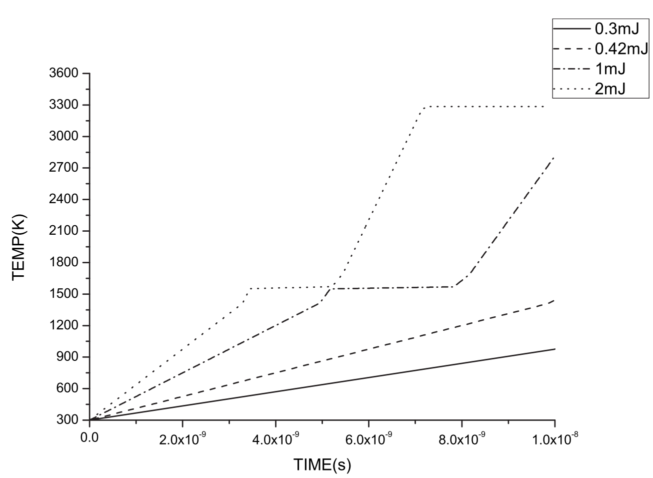

FIG. 3. Temperature evolution of the spot center for 10 ns laser.

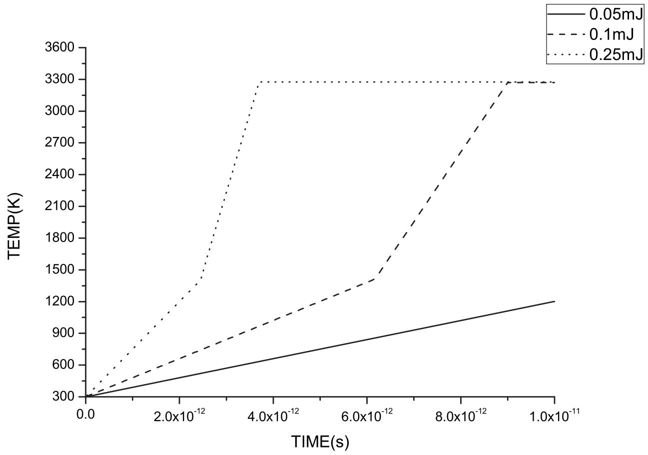

FIG. 4. Temperature evolution of the spot center for 10 ps laser.

when the laser energy is raised to 0.2 J and the corresponding energy density is 159 J/cm^2. Hereby when the temperature reaches the melting point of silicon 1412 K, the damage threshold is about 122 J/cm^2 for 1 ms laser by linear interpolating. For 10 ns laser, as shown in Fig. 3, the values of the peak temperature are 975 and 1444 K when the incident laser energy are 0.3 mJ and 0.42 mJ, and the corresponding energy density are 3.82 J/cm^2 and 5.35 J/cm^2, respectively. Thus, the damage threshold of silicon for 10 ns laser is about 5.22 J/cm^2. The temperature distribution for 10 ps laser is displayed in Fig. 4. The incident laser energy is 0.05 mJ and the corresponding laser energy is 0.63 J/cm^2 which can attain a peak temperature rise up to 1201 K. The maximum temperature reaches the vaporization temperature when the laser energy increase to 0.1 mJ and the corresponding energy density is 1.27 J/cm^2. Thereby, the damage threshold of silicon for 10 ps laser is approximately 0.78 J/cm^2.

Temperature of the spot center rises rapidly and reaches the melting point gradually due to the high thermal conductivity when silicon is irradiated by laser. It is obvious from Table I that the absorption coefficient of silicon in liquid state is much bigger than solid state, therefore, temperature increases sharply in the liquid region as compared to the solid region. Furthermore, the platform as shown in Figs. 2 and 3 represents liquid–solid and vapor–liquid phases change process, respectively. The temperature keeps constant during the liquid–solid and liquid–vapor phases change process no taking account of the phase of silicon after vaporization. In addition, it should be noted here that the platform of solid–liquid phase change is not obvious as presented in Fig. 4, because the temperature rises up sharply and reaches the melting point in so short time for picosecond laser irradiation.

IV. EXPERIMENTAL RESEARCH

A. Experimental set-up description

To confirm the theoretical results and the numerical results, experiments have been performed on single-crystal silicon, which is irradiated by millisecond and nanosecond laser. The experimental arrangement is schematically shown in Fig. 5. The studies about the damage threshold for picosecond laser have been investigated a lot in the past few years. Herein, it should be noted that the experimental result of damage threshold for picosecond laser refers to the previous studies.

A 1064 nm Nd:YAG laser system is applied in the experiment. The long-pulse laser is Melar-50 of Beamtech Co. with a pulse width between 0.5 and 2.5 ms and the maximum output energy is 50 J. The short-pulse laser is Continuum Surelite II-10 with a 10 ns pulse width. The output laser is focused by $f=25$ cm lens L1 and irradiated the surface of sample which is located at the lens focus. The spot radius of 1 ms laser is 200 μm, and the spot radius of 10 ns laser is 50 μm. A He–Ne laser beam is used as a probe beam and focused onto the site to be damaged by lens L2 with dimensions smaller than Nd:YAG spot size. As shown in Fig. 5, the photodiode monitors the reflectivity of the damaged location with an interference filter in front of the detector which only gets across the 532 nm light. The signal is monitored and recorded by oscillograph. The energy meter takes real-time detection of the incident laser energy. The laser energy is 1–20 J and 0.1–250 mJ for long-pulse laser and short-pulse laser, respectively. We adjust the incident laser energy density on the sample by inserting attenuator and splitter in the beam path. The energy density of long-pulse laser and short-pulse laser is varied from 79.6 J/cm^2 to 1592 J/cm^2 and in

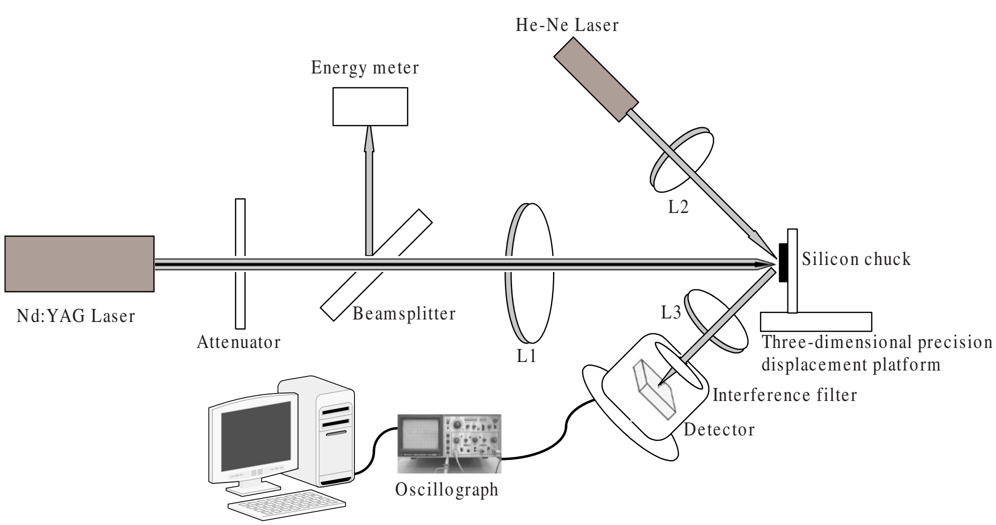

FIG. 5. (Color online) Experiment setup for laser damage.

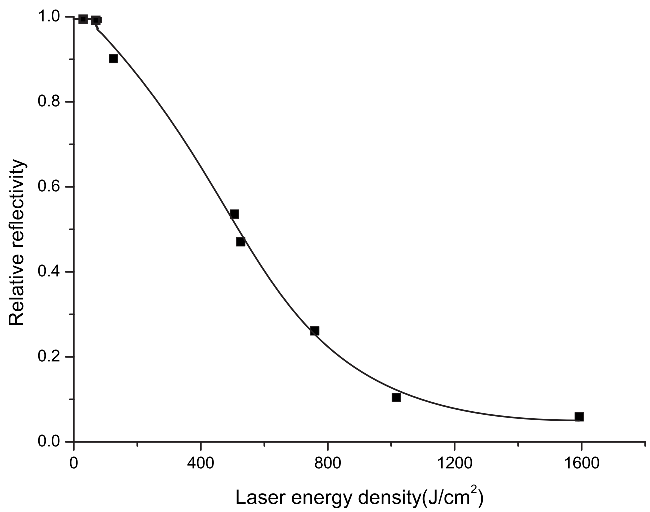

FIG. 6. Relative reflectivity vs laser energy density for 1 ms laser.

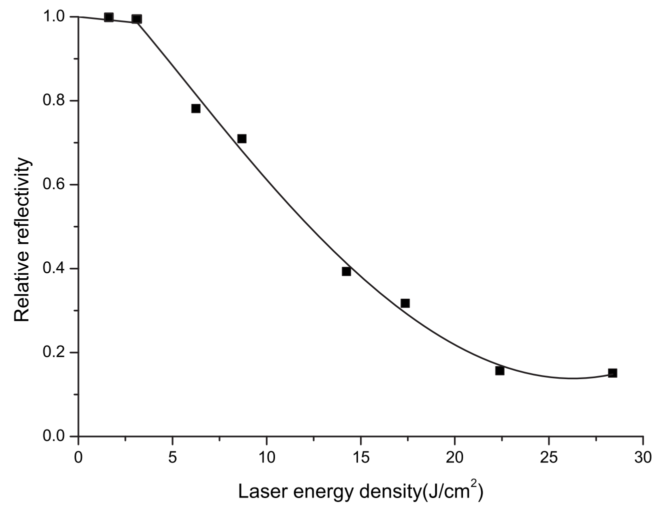

FIG. 7. Relative reflectivity vs laser energy density for 10 ns laser.

the range of 0.127 J/cm^2–318.5 J/cm^2. Under the same laser energy density, the measurements have been repeated ten times to obtain the average of the relative reflectivity which is input into a computer for processing. Therefore, the experimental results of damage threshold for 1 ms and 10 ns laser base on hundreds of damaged spots. The experiment is performed at room temperature in air under normal atmospheric pressure. The sample used is ⟨100⟩ single-crystal silicon. The silicon target is set on three-dimensional precision displacement platform which is initially polished and cleaned in isopropyl alcohol, deionized water, afterwards dipped into dilute HF to remove the oxide layer on its surface and finally is rinsed in the deionized water.

B. Experimental result

The reflectivity of sample surface keeps invariable before the temperature of surface rises up to the melting point and phase change happens. It can be seen from Fig. 6 that the leftward of curve is a little platform. The surface of silicon became rough after melting and the molten crater induce the decrease in probe beam intensity, which cause transient reflectivity declining. The relative reflectivity versus laser energy density for 10 ns laser is displayed in Fig. 7, in which the leftward of curve is platform similarly.

We define the damage threshold as the value of the laser energy density when the relative reflectivity reduces 10%. Thus, it will be seen form the Figs. 6 and 7, that the LIDT of silicon is 127.2 J/cm^2 and 4.8 J/cm^2 for 1 ms and 10 ns pulse width laser, respectively. The experimental result of damage threshold for 10 ps laser is 0.7 J/cm^2.[20,21] The damage thresholds of silicon for three pulse widths are listed in Table II. As depicted in Table II, the damage thresholds obviously increase with the increase in laser pulse width. The increasing rate of the damage threshold of silicon for short-pulse is lower than long-pulse laser obviously, which is accordance with the character of the high instantaneous power intensity and the low energy density for short-pulse laser.

C. Discussion

There are significant differences for LIDT of silicon which may attribute to the different damage mechanisms for different pulse widths laser.

For millisecond laser, energy cumulative effect is noticeable during the long-pulse laser irradiation. The laser peak power intensity is too low to ionize electron and the pulse duration is much longer than the collision time of particles. The laser energy invert into heat energy and the local thermodynamic equilibrium has been established, which give rise to the increase in the working point temperature and transfer to the approach part around the working point in the way of thermal transmittance. Meanwhile, thermal stress induced by thermal expansion is released via thermal softening. Hence, thermal effect is the main damage mechanism for millisecond long-pulse laser.

For nanosecond and picosecond laser, the corresponding laser electric field intensity is high enough and the pulse duration is so short that plenty of electrons accumulate in the conduction band instantly and are excited, which generate air breakdown and plasma formation. Plasma continues to absorb laser energy rapidly, reach superheated state and expand sharply. Shock wave and plasma spark are developed which can induce the impact and mechanical damage of the

TABLE II. The damage threshold of silicon for three pulse widths laser.

Laser pulse width	Theoretical result	Numerical result	Experimental result
1 ms	124 J/cm^2	122 J/cm^2	127.2 J/cm^2
10 ns	5 J/cm^2	5.22 J/cm^2	4.8 J/cm^2
10 ps	0.73 J/cm^2	0.78 J/cm^2	0.7 J/cm^2 [a]

[a]Reference 20.

silicon surface. Meanwhile, the ionized materials are ejected outside. In a word, it is attributed to the thermal damage, shock damage and mechanism damage for nanosecond and picosecond pulse width laser.

V. CONCLUSION

In summary, it is shown that the LIDT of silicon is 127.2, 4.8, and 0.7 J/cm^2 for 1 ms, 10 ns, and 10 ps laser experimentally. The experiment result is in reasonable agreement with our theoretical calculations and the numerical results, which proved the rationality of our theoretical and numerical models. The results show that the damage threshold of material increases obviously with the increasing of laser pulse width, which indicates that the damage mechanism of laser–silicon interaction for long-pulse laser is different from the short-pulse laser.

Results in our study can provide theoretic foundation for the investigation on laser–silicon interaction mechanism and the evaluation of the heat effect for different pulse widths.

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